

Single-cell transcriptomic analysis reveals CAF heterogeneity and signaling programs in breast cancer

Esraa Mohamed

202301162

Cairo

Bioinformatics & computational
biology unit

s-esraa.mustafa@zewailcity.edu.eg

Shimaa Mohamed

202301854

Cairo

Bioinformatics & computational
biology unit

s-shimaa.abdelhalim@zewailcity.edu.eg

Menatalla Essam

202300497

Cairo

Bioinformatics & computational
biology unit

s-menatalla.nashat@zewailcity.edu.eg

INTRODUCTION/MOTIVATION

Breast cancer consists of multiple subtypes and can involve different kinds of tumor cells and microenvironments. Breast cancer heterogeneity is clinically important. It can alter the response of tumors to therapies, the aggressiveness and outcome of the disease, and the course of the tumor. Analyzing breast cancer at the single-cell level is important as traditional methods group cells based on the average of their expression, and therefore lose information from the subpopulations. Wu et al. demonstrated that the diversity of the breast cancer tumor and its components could be assessed using single-cell transcriptomics within the tumor microenvironment along with spatial tissue analysis.

The differences between ER+ and triple-negative breast tumors were analyzed. ER-positive tumors contain tumor cells with estrogen receptors and exhibit proliferation due to the hormone. In contrast, triple-negative tumors do not possess any of the three key proteins, namely estrogen receptors, progesterone receptors, and the HER2 receptor. The triple negative phenotype has been shown to be more clinically important compared to ER+ tumors because they grow and spread more, are more aggressive, and have more limited treatment options. Triple-negative breast cancer is associated with a lower prognosis than ER+ breast cancer due to the more limited treatment options.

1. PROBLEM DEFINITION

The main issue examined in this research project is that bulk RNA-seq fails to explain breast cancer heterogeneity. This is because bulk RNA-seq provides an average gene expression profile that is reflective of a diverse mixture of cellular constituents and can therefore obfuscate important rare cell populations and subtype-specific expression patterns.

The primary biological question of this project is, at the single-cell level, how do ER+ and TNBC breast cancer samples differ in their cellular constituents and gene expression? The primary computational challenge is to handle high-dimensional single-cell RNA-seq data in order to perform the necessary quality control, normalization, and clustering, and finally, data visualization and differential expression analyses to elucidate the cellular components of each of the breast cancer types.

2. RELATED WORK

Recently, scRNA-seq has become an advanced technology that allows studying tumor heterogeneity at single-cell resolution. In particular, scRNA-seq reveals different types of cells present in the breast tumor tissue, such as malignant, immune cells, and various types of stroma, for instance, cancer-associated fibroblasts (CAFs). CAFs are involved in numerous processes, such as extracellular matrix remodeling, immune regulation, metastasis formation, and the development of therapy resistance [7]. There exist several algorithms designed for scRNA-seq data analysis. The most popular software solution is the package from the official Seurat website, which includes quality control, normalization, dimensionality reduction, clustering, differential expression analysis, and cell type annotation [8], [9].

Seurat uses PCA followed by graph-based clustering and visualization via UMAP to cluster cells based on their transcription profiles. A significant problem associated with scRNA-seq analysis is batch correction because samples may differ in terms of patient identity, experimental procedures, etc. In order to correct these differences, Harmony was proposed, a fast and reliable algorithm for batch effect correction that does not destroy biological information [10].

Harmony gained popularity in tumor microenvironment studies because of its effectiveness in multi-sample integration without over-correction. Specifically, for breast cancer, Wu et al. built a complete single-cell atlas of human breast tumors based on the GSE176078 data set, highlighting significant heterogeneity in all three major compartments, namely epithelium, immunity, and stroma [7]. In this way, they found several distinct fibroblast states characterized by specific extracellular matrix, inflammation, and immune features. Furthermore, recent studies have investigated CAF heterogeneity by defining biologically distinct subtypes, including myofibroblastic CAFs, inflammatory CAFs, matrix CAFs, and antigen-presenting CAFs, defined by a group of canonical genes, such as ACTA2, CXCL12, COL1A1, and HLA-DRA [11].

Different CAF states play different roles in tumor development, fibrosis, immune modulation, and drug resistance. Finally, cell-cell communication analysis is another approach that is widely used in tumor microenvironment research. CellChat is a tool that allows inference of intercellular signaling based on existing ligand-receptor interactions between cells [12]. Modeling cell-to-cell communication probability makes it possible to identify key

communication networks in CAF subsets and other tumor-associated cells. Thus, the workflow adopted in this project integrates well-established and innovative approaches, such as Seurat clustering and annotation, Harmony correction for batch effect, identification of distinct CAF subtypes, and cell-cell communication analysis by CellChat.

3. PROPOSED METHODS

3.1 Intuition

Seurat provides a solid framework for single-cell preprocessing, clustering, and visualization. Harmony was used to reduce batch effects across samples, while CellChat enabled inference of ligand–receptor communication among CAF subtypes. Together, these methods allow comprehensive characterization of CAF heterogeneity and signaling interactions in breast cancer.

3.2 Preprocessing

This project utilized a Seurat-based single-cell RNA-seq workflow to analyze the breast cancer dataset GSE176078 and compare ER+ versus TNBC samples. Seurat was chosen for this project as it is a complete and very popular workflow for handling scRNA-seq data including quality assessment and control, normalization, dimensionality reduction, clustering and visualization, as well as differential analysis.[4]

This workflow was used as breast cancer is a very heterogeneous disease, single-cell RNA-seq has the ability to uncover the diversity of tumor, immune and stromal cells which may be hidden in bulk RNA-seq averages. GSE176078 breast cancer atlas demonstrates that single-cell analysis can capture cellular architecture in addition to the sub-type associated variety in human breast tumors.[5]

The workflow begins by loading the count matrix and associated metadata for the creation of a Seurat object. Quality control is performed to remove cells that are of a low quality. Once this is completed, normalization is performed to make the expression of the genes across the cells more comparable. Genes that are most variable are selected as they are most likely to capture the greatest biological differences that underscore cell population diversity. A PCA is performed for the purpose of dimensionality reduction. For the purpose of visualization, UMAP is employed.

Clustering is then used to group cells that are similar in a transcriptional manner. Lastly, differential expression analysis is performed to ascertain the genes that are expressed in either ER+ or TNBC samples. Marker detection and differential expression analysis for comparison of clusters or subgroups are also supported by the Seurat package.[4]

The comparison of ER+ and TNBC samples will be more valid if batch variation among the samples is accounted for. In such a scenario, the harmony package can be used post PCA to alleviate some of the variation while retaining biological relevance.[6]

3.3 Annotation

Cluster Annotation and Validation

The identified cell clusters were annotated based on metadata labels provided in the dataset and the expression of canonical marker genes. Major breast cancer cell populations, including immune cells, epithelial cells, endothelial cells, and cancer-

associated fibroblasts (CAFs), were identified using the available metadata and gene-expression patterns.

To validate fibroblast-related clusters, visualization analyses including DimPlot, FeaturePlot, and marker-expression comparisons were performed. Canonical fibroblast markers such as COL1A1, COL1A2, DCN, LUM, FAP, PDGFRA, and PDGFRB were examined because these genes are widely associated with stromal fibroblast populations in breast cancer tumor microenvironments.

Marker-based annotation is considered a commonly used and biologically valid strategy in exploratory single-cell RNA sequencing studies. However, because manual annotation may introduce interpretation bias, the annotation process was supported using both dataset metadata and previously reported CAF marker signatures from published breast cancer studies.

CAF Population Extraction and Re-Clustering

After annotation, CAF populations were extracted from the complete breast cancer dataset using metadata-based filtering. The extracted CAF subset was subsequently reprocessed independently through normalization, highly variable gene selection, scaling, principal component analysis (PCA), UMAP visualization, and graph-based clustering.

Secondary re-clustering of CAFs is considered an important analytical step because fibroblast populations exhibit substantial transcriptional heterogeneity and may contain multiple functional states that are not fully distinguishable during whole-tumor clustering.

Recent breast cancer scRNA-seq studies have demonstrated that stromal re-clustering improves the resolution of CAF subtype diversity and facilitates identification of biologically meaningful subpopulations associated with metastasis, extracellular matrix remodeling, inflammatory signaling, and immune regulation [17]

The CAF-specific UMAP visualization revealed transcriptionally distinct fibroblast populations, supporting the presence of substantial heterogeneity within CAF populations in breast cancer samples.

Marker Gene Identification and Biological Interpretation

Marker genes for each CAF cluster were identified using the FindAllMarkers function implemented in Seurat with thresholds for minimum percentage expression (min.pct = 0.25) and log fold-change (logfc.threshold = 0.25). The analysis was restricted to positively enriched genes (only.pos = TRUE) in order to identify genes highly expressed within each CAF cluster relative to the remaining CAF populations.

To biologically interpret CAF heterogeneity, canonical fibroblast markers and previously reported CAF subtype markers from breast cancer studies were manually examined across clusters. Visualization approaches including

FeaturePlot, DotPlot, and heatmaps were used to compare subtype-specific gene-expression patterns among CAF clusters.

CAF subtype annotation was therefore performed manually based on known biological marker signatures rather than automated reference-based annotation methods. Clusters enriched in ACTA2, TAGLN, MYL9, and TPM2 expression were annotated as myofibroblastic CAFs (myCAFs). Clusters showing elevated expression of inflammatory cytokines and chemokines including CXCL12, CXCL14, IL6, and CCL2 were annotated as inflammatory CAFs (iCAFs). In contrast, clusters characterized by strong extracellular matrix-associated gene expression such as COL1A1, COL3A1, FN1, and POSTN were classified as matrix-associated CAFs (matrixCAFs).

The identified CAF subtypes are consistent with previously reported stromal populations in breast cancer single-cell studies, where myCAFs are associated with extracellular matrix remodeling and contractile activity, whereas iCAFs are linked to inflammatory signaling and immune modulation within the tumor microenvironment[13, 14].

3.4 Cell chat

The CAF Seurat object, resulting from CAF extraction and reclustering, was used for CellChat analysis. The CellChat object was first created using the normalized gene expression matrix and the CAF subtype information generated from the Seurat pipeline. The refined CellChatDB.human database containing experimentally validated ligand-receptor interactions and signaling pathways related to human cell-cell communication was imported into CellChat. Using the refined database, signaling communications between CAF populations were characterized in a biologically relevant way.

To focus on communications-associated genes only, the dataset was subsetted using subsetData() that only selects signaling-associated genes with ligand-receptor interactions from the whole gene list. Then, overexpressed communication genes within CAF populations were identified by identifyOverExpressedGenes(), and significant ligand-receptor pairs between CAF populations were revealed by identifyOverExpressedInteractions().

Subsequently, communication probabilities between CAF populations were computed by computeCommunProb(). Using it, communication between two cell populations was characterized by how expressed ligands are from one cell type and how expressed receptors are from another cell type.

Pathway-level communications between cell populations were also analyzed using computeCommunProbPathway(), in which several ligand-receptor pairs were mapped into biologically meaningful signaling pathways.

To reveal the overall communication profiles between cell types, aggregated signaling interactions were demonstrated as a network using aggregateNet(). Lastly, the signaling networks and communications between CAF cell types were visually explored with circle plots that show the networks of cell-cell communication, bubble plots showing the ligand-receptor interactions between specific cell populations, and pathway rank plots that show which signaling pathways are most active and signaling role heatmaps that compare outgoing and incoming signaling activities among cell types.

4. EVALUATION/EXPERIMENTS/RESULTS

4.1 Dataset

In our project, we utilized the publicly available breast cancer scRNA-seq dataset GSE176078 obtained from: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE176078>

The dataset contains single-cell transcriptomic profiles generated from 26 samples of primary human breast cancer tumors using the 10x Genomics platform. Expression matrices were provided in sparse matrix format, including:

- matrix.mtx
- barcodes.tsv
- genes.tsv
- Metadata.csv

Associated metadata containing sample and cell-type annotations were also utilized during downstream analysis and validation. The dataset includes multiple cellular populations within the primary breast tumor microenvironment, including malignant epithelial cells, immune cells, endothelial cells, and stromal fibroblasts.

4.2 List of questions

- Can scRNA-seq analysis identify distinct cellular populations within the breast cancer tumor microenvironment?
- Are cancer-associated fibroblasts (CAFs) heterogeneous, and can they be classified into biologically distinct CAF subtypes?
- Which marker genes characterize each CAF subtype?
- What molecular differences exist between TNBC and ER+ breast cancer samples?
- How do CAF subtypes communicate with each other through ligand-receptor interactions?
- Which signaling pathways are most active among CAF populations in the breast cancer microenvironment?
- Which CAF subtypes exhibit the strongest signaling and communication activities within the tumor microenvironment?

4.3 Results and discussion

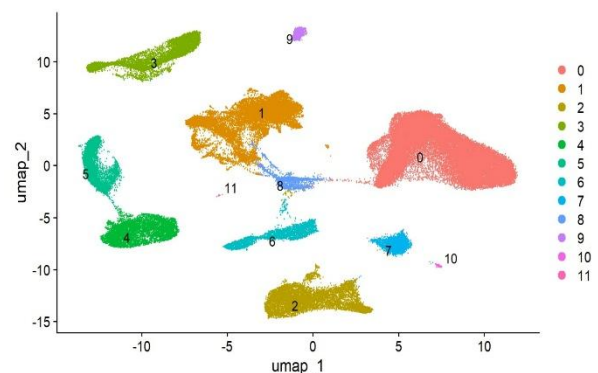


Figure 1. UMAP visualization of clustered breast cancer single-cell RNA-seq data.

Every data point corresponds to a single cell taken from the GSE176078 breast cancer dataset, with cells located closer being

more alike in terms of gene expression pattern. The figure demonstrates 12 clusters with distinct transcription profiles numbered from 0 to 11, thereby proving that the pre-processing approach has resulted in clustering of the cells into different biological groups. It helps to achieve the goal of the study since GSE176078 is dedicated to the investigation of cellular architecture and heterogeneity of human breast cancers. The distinction between several clusters proves the diversity of cells included in the dataset.

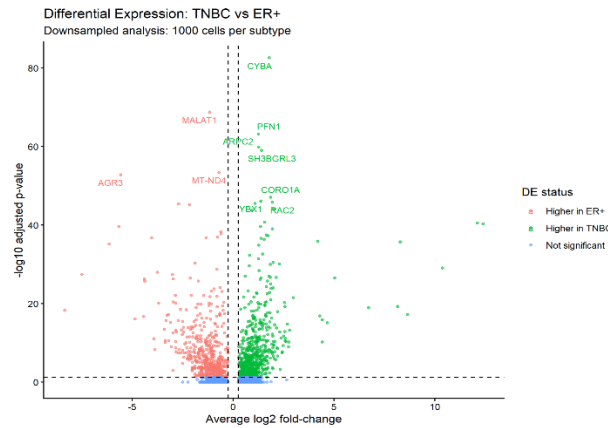


Figure 2. Volcano plot of differential gene expression between TNBC and ER+ breast cancer samples.

The point denotes one gene downsampled to 1000 cells per subtype. The horizontal axis indicates the average log2 fold change, whereas the vertical axis is the $-\log_{10}$ adjusted p-value. The right genes are up-regulated in TNBC, the left genes are up-regulated in ER+, and the blue genes are not significant. As evident from the plot, there are some significantly upregulated genes in TNBC, including CYBA, PFN1, SH3BGR3, CORO1A, RAC2, and ARPC2. Genes like AGR3, MALAT1, and MT-ND4 are significantly upregulated in ER+. It is an important Seurat feature called differential expression analysis, which is done to find significantly differentially expressed genes between clusters or biological groups. This result implies that there are different expressions of genes in ER+ and TNBC patients.

The proposed Seurat-based workflow has several advantages. First and foremost, it is intended for single-cell RNA-seq as it consolidates quality control, normalization, dimensionality reduction, clustering, and visualization, as well as differential expression in one single workflow. Second, if compared to bulk RNA-seq, it is better for this project as it retains cell-level data. Considering this, it becomes possible to visualize and compare the constituent, as opposed to averaged, cell populations of breast cancer. The GSE176078 atlas demonstrates that single-cell methods are invaluable for elucidating not just cellular diversity in breast cancer, but also the associated subtype-specific heterogeneity. Third, in cases where there are batch effects across samples, Harmony can be implemented after PCA to help reduce dataset-specific variation while preserving biologically meaningful cell structure.

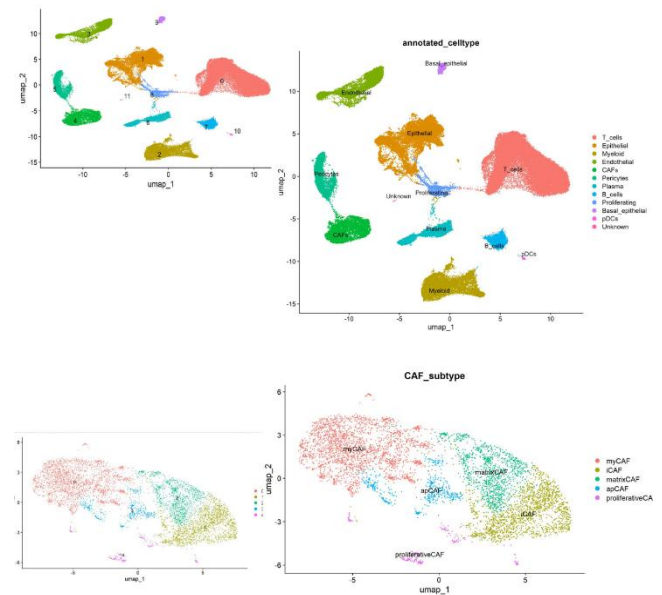
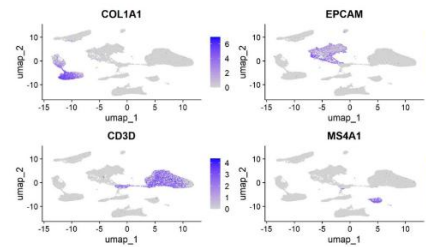


Figure 3, shows cell type markers using Umap.

Regarding Figure 3, and from the first visualization of the breast cancer data by UMAP, we can clearly see that there are 12 transcriptionally different clusters in this tissue sample, suggesting high diversity among cells in the tumor microenvironment. The clusters correspond to T cells, epithelial cells, myeloid cells, endothelial cells, pericytes, plasma cells, B cells, proliferating cells, pDCs, basal epithelial cells, and CAFs. The clear separation of these populations on the plot demonstrates that the workflow, including preprocessing, normalization, PCA, and clustering, is successful in extracting transcriptional variation among the samples. Specifically, the CAF cluster forms a rather isolated cluster compared to other immune and epithelial populations. This allows us to conclude that the extraction of fibroblasts was correct for further analyses[17].

[1] Feature Plot



[2] Dpt Plot

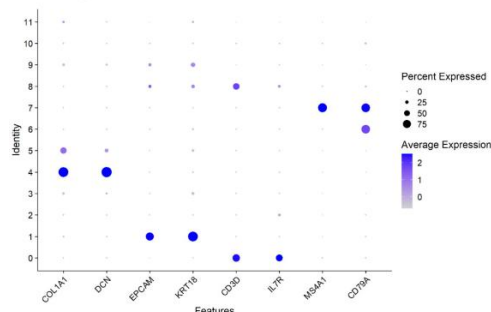


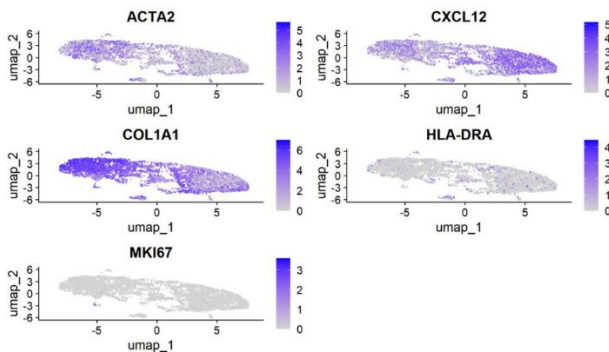
Figure 4 shows cell-type markers using a feature plot and a dot plot

As shown in Figure 4, FeaturePlot shows the expression levels of canonical markers of various cell types in the clusters. High levels of COL1A1 are mainly found in the cluster predicted as CAF, confirming the identification of fibroblasts. At the same time, strong expression of EPCAM is observed in epithelial clusters, expression of CD3D indicates T cells, and the presence of MS4A1 corresponds to the B cell populations. These results coincide with previous breast cancer scRNA-seq data analysis [18]

Additionally, the DotPlot visualization provided evidence for cluster-specific marker enrichment. Larger and highly saturated colored dots represented higher mean gene expression and higher proportions of expressing cells among the respective clusters, validating the correctness of cluster annotation[15].

CAF Subtypes Markers

[1] Feature Plot



[2] Dpt Plot

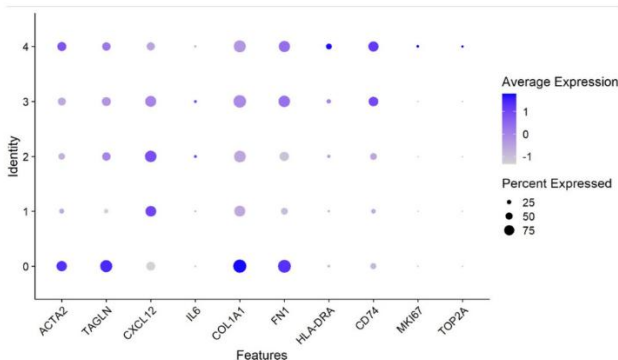


Figure 5, shows Clustering of CAFs Highlights Fibroblast Phenotypic Diversity, using a feature and a dot plot.

In the course of re-clustering of CAFs after their extraction, multiple subpopulations of CAFs were identified. The CAF-specific UMAP highlighted different fibroblast states, emphasizing fibroblast phenotypic diversity[16, 17].

Identified CAF subtypes:

myCAFs (myofibroblastic CAFs)

Defined by upregulation of ACTA2 and TAGLN, genes associated with contraction and extracellular matrix remodeling.

iCAFs (inflammatory CAFs)

Defined by upregulation of several pro-inflammatory cytokines such as CXCL12[16].

matrixCAFs

Exhibit strong extracellular matrix-related signature including COL1A1 and FN1, implying active involvement in stromal remodeling and collagen production[17].

apCAFs and proliferative CAFs

Less numerous subgroups with a distinctive transcription profile. Proliferative CAFs do not show any evidence of proliferation-related gene expression (MKI67, TOP2A)[18].

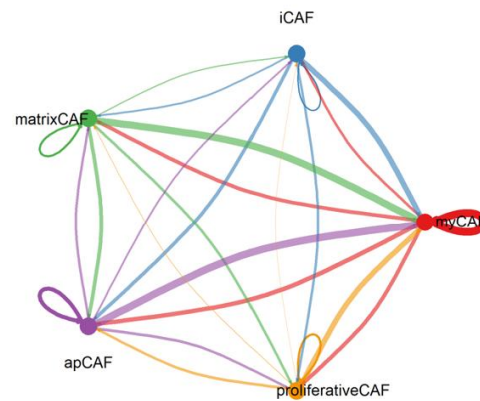


Figure 6. CellChat communication network between CAF subtypes.

Depending on Figure 6, the ligand-receptor interactions identified using Cell Chat reveal extensive signaling networks among CAF subtypes; both myCAF and apCAF showed significantly higher outbound signalling the circle plot figure, the thickness of the arrow represents the strength of the signals send from one cluster to another, and it is not obvious because of the zoom deficiency, but the there is an pointer on the arrow representing which cluster is the signal sender and which is the receiver.

This can suggest that both myCAF and apCAF might play important roles in the microenvironment.

This method of representation is good because it clearly shows which cluster talks to which cluster. On the other hand, it does not show the name of the signal.

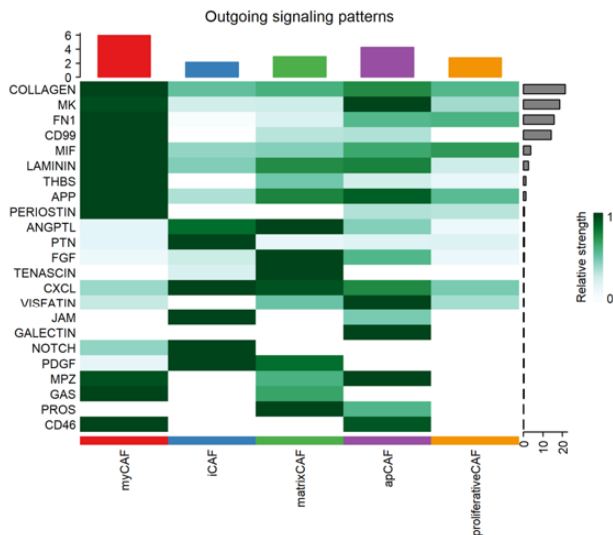


Figure 7. Heatmap showing signaling pathway activities across CAF subtypes.

As shown in Figure 7, among CAF populations, COLLAGEN, MK, and FN1 have shown highly active pathways with CellChat analysis. Previous studies on breast cancer indicated these pathways are associated with EMT, fibroblast activation, tumor invasion and metastasis, and immune regulation, which supports the biological meaning and validates the CellChat results. Moreover, by viewing a heatmap of the signaling roles across different CAF subtypes, we can clearly illustrate the heterogeneous signaling abilities among subtypes within the microenvironment.

This method of showing the results (Figure 7) is considered the most helpful method as it shows the depth of the process and the signals among the different types of CAFs and their strengths.

4.4 Limitations

4.4.1 limitations in preprocessing

The limitations of the workflow in question are as follows. The quality of the outcome relies on the quality of the data input, the QC thresholds selected by the user, the clustering resolution, and the number of cells included in the analysis. In the volcano plot, forensic downsampling to 1000 cells per subtype makes the ER+ vs TNBC comparison more balanced; however, it is likely to capture less biological variation when compared with the full dataset. Finally, as primarily a visualization technique, UMAP is limited in the sense that the clusters made in the UMAP should be further biologically validated using known marker genes or via additional annotation. Thus, while the workflow in question is quite robust in terms of finding cell populations and cell populations, and providing lists of genes that are differentially expressed, it must be interpreted in conjunction with validation of markers, pathway analysis, and biological literature.

4.4.2 Limitations in annotation

4.4.2.1 Possibility of Over-Interpretation in Manual Annotation of CAF Subtypes

Subtypes for CAFs were annotated manually based on their canonical marker genes. Although manual annotation is common practice in exploratory scRNA-seq analyses, CAF biology is quite dynamic and continuous in nature [20].

For instance:

- The COL1A1 gene was observed in more than one cluster.
- The CXCL12 gene was not solely expressed within a single cluster.
- There was an overlap between the gene expression profile of some CAF subtypes.

Thus, the CAF subtype labels should be considered dominant transcriptional states rather than strict biological phenotypes.

4.4.2.2 Low Sensitivity in Identifying Rarer CAF Populations

The apCAF and proliferativeCAF populations appeared to be less abundant than other major CAF populations. This may be due to:

low sequencing depth,

limited number of cells,

or low resolution during the clustering step.

Rarer stromal cell populations can sometimes be challenging to detect in standard scRNA-seq experiments because of their smooth transcriptional transition states.

4.4.2.3 Absence of a Doublet Identification Process

No doublet detection method was used prior to clustering.

This is significant because doublets have the ability to produce spurious hybrid transcriptional profiles that resemble intermediate cell types. This means that an apparent intermediate state between fibroblast and immune cells could be due to doublets, not biological intermediates.

Current approaches to scRNA-seq data analyses usually include:

DoubletFinder, Scrublet, and scDbfFinder

prior to clustering [21].

4.4.2.4 Dependency on UMAP

This process was reliant upon separation by UMAP. But since UMAP is a nonlinear dimensionality reduction approach mainly used for visualization purposes, it does not always ensure that biological distance is preserved.

This means that separated clusters are not always different biologically, and overlapping clusters are not always similar biologically.

More statistical tests would improve this study [22].

4.4.3 limitations in Cell chat

- CellChat analysis is heavily reliant on the annotated databases of ligands and receptors, as well as computational prediction methods.
- Identified signaling interactions have not been experimentally verified.
- The analysis was done only with primary breast cancer specimens, with no metastatic specimens included.
- Cell-cell interactions are lost when spatial data is ignored.
- The communication probability will likely be affected by genes with low expression levels or very rare cell types.

4.5 Evaluation

The COLLAGEN, MK and FN1 signaling pathways were the most active pathways identified in CAF populations. In order to support the biological significance of the analyzed CellChat results, the data

from published breast cancer literature were compared and were found to be in agreement with the signaling communication information received from the analysis performed.

Previous breast cancer studies showed that COLLAGEN signaling has a key role in the process of extracellular matrix reorganization, tumor rigidity, invasion, and metastasis [23]. Moreover, FN1 signaling was found to have a crucial role in the activation of fibroblasts, tumor growth, and metastatic potential in the breast cancer tumor microenvironment [24]. Finally, MK signaling was reported to have a role in tumor growth, neovascularization, and immunomodulation in breast cancer [25].

The convergence of CellChat analysis with previously published findings in breast cancer further supports the biological validity of the identified communication networks and also highlights the potential key roles played by the detected CAF signaling communications in breast cancer progression.

5. CONCLUSION

In this project, we successfully used scRNA-seq analysis to characterize the cellular diversity of the breast cancer tumor microenvironment. We performed data preprocessing, clustering, and annotation analysis, and identified multiple key cell populations (e.g., epithelial cells, immune cells, endothelial cells, and cancer-associated fibroblasts, CAFs). Based on the re-clustering of CAFs, we uncovered diverse transcriptionally distinct CAF subpopulations (e.g. MyCAF, iCAF, matrixCAF, apCAF, proliferativeCAF). And these findings support the diverse nature of CAFs in breast cancer.

Via CellChat analysis, we found wide spread of ligand-receptor pairs among CAF subtypes within the tumor microenvironment. Furthermore, outgoing signal activity of myCAF and apCAF was higher than the other CAF subtypes; these might be the main signal mediators within breast cancer tumors. Through analysis on pathways, COLLAGEN, MK, and FN1 are some of the most active signaling pathways among CAFs. These pathways are known in many studies on breast cancer that strongly correlate with extracellular matrix remodeling, fibroblast activation, tumor invasion/metastasis, and immune regulation.

Thus, our findings from CellChat analysis are consistent with previously reported breast cancer mechanisms and support the biological validity of the communication pathways and signaling pathways we have discovered. In addition, cell-cell communication analysis based on signaling role heatmap analysis revealed the heterogeneous signaling activities of CAFs, indicating diverse functional identities between the CAF populations in the tumor microenvironment.

All results indicate the crucial roles of communication networks among CAFs in breast cancer progression, and the power of scRNA-seq combined with CellChat analysis to investigate tumor microenvironment interaction

6. FUTURE WORK

Further work can include the integration of spatial transcriptomics data to better analyze the spatial organization of CAF communication in the tumor microenvironment. The addition of metastatic breast cancer samples and a larger dataset could offer more detailed analyses of CAF heterogeneity and signal dynamics at various stages of breast cancer.

Experimental validation of the proposed ligand-receptor pairs and signaling pathways will further reinforce the biological

interpretation of CellChat results. Meanwhile, additional computational methods and integration of multi-omics would provide more profound insights into CAF-mediated tumor progression and immune regulation.

7. REFERENCES

- [1] S. Z. Wu et al., “A single-cell and spatially resolved atlas of human breast cancers,” *Nature Genetics*, vol. 53, no. 9, pp. 1334–1347, Sep. 2021, doi: 10.1038/s41588-021-00911-1.
- [2] “NCI Dictionary of Cancer Terms,” Cancer.gov. <https://www.cancer.gov/publications/dictionaries/cancer-terms/def/triple-negative-breast-cancer>
- [3] Y. M. Hawsawi et al., “Recent progress and applications of single-cell sequencing technology in breast cancer,” *Frontiers in Genetics*, vol. 15, p. 1417415, Sep. 2024, doi: 10.3389/fgene.2024.1417415.
- [4] “Seurat - Guided Clustering Tutorial.” https://satijalab.org/seurat/articles/pbmc3k_tutorial.html
- [5] “GEO Accession viewer.” <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE176078>
- [6] I. Korsunsky et al., “Fast, sensitive and accurate integration of single-cell data with Harmony,” *Nature Methods*, vol. 16, no. 12, pp. 1289–1296, Nov. 2019, doi: 10.1038/s41592-019-0619-0.
- [7] A single-cell and spatially resolved atlas of human breast cancers S. Z. Wu *et al.*, “A single-cell and spatially resolved atlas of human breast cancers,” *Nature Genetics*, vol. 53, no. 9, pp. 1334–1347, 2021.
- [8] Comprehensive Integration of Single-Cell Data T. Stuart *et al.*, “Comprehensive Integration of Single-Cell Data,” *Cell*, vol. 177, no. 7, pp. 1888–1902.e21, 2019.
- [9] Integrated analysis of multimodal single-cell data Y. Hao *et al.*, “Integrated analysis of multimodal single-cell data,” *Cell*, vol. 184, no. 13, pp. 3573–3587.e29, 2021.
- [10] Harmony: fast, sensitive and accurate integration of single-cell data I. Korsunsky *et al.*, “Harmony: fast, sensitive and accurate integration of single-cell data,” *Nature Methods*, vol. 16, no. 12, pp. 1289–1296, 2019.
- [11] Comprehensive characterization of CAF heterogeneity in breast cancer by single-cell RNA sequencing X. Chen *et al.*, “Comprehensive characterization of CAF heterogeneity in breast cancer by single-cell RNA sequencing,” *Frontiers in Oncology*, vol. 13, 2023.
- [12] Inference and analysis of cell-cell communication using CellChat S. Jin *et al.*, “Inference and analysis of cell-cell

- communication using CellChat,” *Nature Communications*, vol. 12, no. 1, 2021.
- [13] Y. Xu, P. Lin, Y. Zhu, Q. Zhang, and J. Zhou, “Applying integrated transcriptome and single-cell sequencing analysis to develop a prognostic signature based on M2-like tumor-associated macrophages for breast cancer,” *Discover Oncology*, vol. 16, no. 1, Mar. 2025, doi: <https://doi.org/10.1007/s12672-025-02161-7>.
- [14] L. Zhang, Q. Zhang, M. Yang, M. Xu, Q. Li, and Q. Chen, “Integrating single-cell and bulk transcriptomes with machine learning reveals a CAF signature for immunotherapy response in dMMR endometrial cancer,” *Discover Oncology*, vol. 17, no. 1, Jan. 2026, doi: <https://doi.org/10.1007/s12672-026-04414-5>.
- [15] Zhang S, Li X, Lin J, Lin Q, Wong KC. Review of single-cell RNA-seq data clustering for cell-type identification and characterization. *RNA*. 2023;29(5):517–530.
- [16] Deng H, Wen C, Jiang S, Yu Y, Zhao J, Zhang B. Single-cell analysis reveals one cancer-associated fibroblasts subtype linked to metastasis in breast cancer: MXRA5 as a potential novel marker for prognosis. *American Journal of Cancer Research*. 2024;14(2):526–544.
- [17] Ning L, Quan C, Wang Y, Wu Z, Yuan P, Xie N. scRNA-seq characterizing the heterogeneity of fibroblasts in breast cancer reveals a novel subtype SFRP4+ CAF that inhibits migration and predicts prognosis. *Frontiers in Oncology*. 2024;14:1348299.
- [18] Luecken MD, Theis FJ. Current best practices in single-cell RNA-seq analysis: a tutorial. *Molecular Systems Biology*. 2019;15(6):e8746.
- [19] Elyada E, Bolisetty M, Laise P, et al. Cross-species single-cell analysis of pancreatic ductal adenocarcinoma reveals antigen-presenting cancer-associated fibroblasts. *Cancer Discovery*. 2019;9(8):1102–1123.
- [20] Öhlund D, Elyada E, Tuveson D. Fibroblast heterogeneity in the cancer wound. *Journal of Experimental Medicine*. 2017;214(3):579–596.
- [21] McGinnis CS, Murrow LM, Gartner ZJ. DoubletFinder: doublet detection in single-cell RNA sequencing data using artificial nearest neighbors. *Cell Systems*. 2019;8(4):329–337.
- [22] Becht E, McInnes L, Healy J, et al. Dimensionality reduction for visualizing single-cell data using UMAP. *Nature Biotechnology*. 2019;37:38–44.
- [23] M. K. Jena and J. Janjanam, “Role of extracellular matrix in breast cancer development: a brief update,” *F1000Research*, vol. 7, p. 274, Jun. 2018, doi: <https://doi.org/10.12688/f1000research.14133.2>.
- [24] Karin Wang, Fei Wu, Bo Ri Seo, Claudia Fischbach, Weisi Chen, Lauren Hsu, and Delphine Gourdon. 2017. Breast cancer cells alter the dynamics of stromal fibronectin-collagen interactions. *Matrix Biology* 60–61, 86–95. <https://doi.org/10.1016/j.matbio.2016.08.001>
- [25] W. M. Hadida, D. A. Ali, A. A. ELnemr, and H. E. Nofal, “The Role of Midkine in Breast Cancer,” *The Egyptian Journal of Hospital Medicine*, vol. 77, no. 3, pp. 5089–5095, Oct. 2019, doi: <https://doi.org/10.21608/ejhm.2019.49550>.
- [26] “Analysis, visualization, and integration of Visium HD spatial datasets with Seurat,” *Satijalab.org*, 2023. https://satijalab.org/seurat/articles/seurat5_integration
- [27] “GitHub & BitBucket HTML Preview,” *Github.io*, 2026. <https://htmlpreview.github.io/?https://github.com/sqjin/CellChat/blob/master/tutorial/CellChat-vignette.html#create-a-cellchat-object> (accessed May 06, 2026).
- [28] M. D. Luecken and F. J. Theis, “Current best practices in single-cell RNA-seq analysis: a tutorial,” *Molecular Systems Biology*, vol. 15, no. 6, p. e8746, Jun. 2019, doi: 10.15252/msb.20188746.